

CS23334 Fundamentals of Data Science Lab

Experiment 8

Conduct Z-Test for Given Data

Objective: To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

- 1 **Null Hypothesis (H_0):** The average weight of the birds is 150 grams.
- 2 **Alternative Hypothesis (H_1):** The average weight of the birds is not 150 grams.
- 3 **Sample:** Measure the weights of 30 birds randomly selected from the population.
- 4 **Z-Test:** Conduct a Z-test to compare the sample mean to 150 grams.
- 5 **Decision Rule:** Use a significance level of $\alpha = 0.05$.

Code:

```
import numpy as np
import scipy.stats as stats

sample_data = np.array([
    152, 148, 151, 149, 147, 153, 150, 148, 152, 149,
    151, 150, 149, 152, 151, 148, 150, 152, 149, 150,
    148, 153, 151, 150, 149, 152, 148, 151, 150, 153
])

population_mean = 150

sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 150 grams")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weight")
```

OUTPUT:

Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from